

Experiment No.7

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Subject: Deep Learning

Problem statement:

Design the architecture and implement the autoencoder model for Image Compression.

Source Code:

Image Compression with Autoencoder

Import libraries

import tensorflow as tf

from tensorflow.keras import layers, models

import matplotlib.pyplot as plt

import numpy as np

Load MNIST dataset

(x_train, _), (x_test, _) = tf.keras.datasets.mnist.load_data()

Normalize data

x_train = x_train / 255.0

x_test = x_test / 255.0

Flatten images

x_train = x_train.reshape(-1, 784)

x_test = x_test.reshape(-1, 784)

Build Autoencoder Model

Encoder

```

input_img = layers.Input(shape=(784,))
encoded = layers.Dense(128, activation='relu')(input_img)
encoded = layers.Dense(64, activation='relu')(encoded)
bottleneck = layers.Dense(32, activation='relu')(encoded)

# Decoder
decoded = layers.Dense(64, activation='relu')(bottleneck)
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decoded = layers.Dense(128, activation='relu')(decoded)
output_img = layers.Dense(784, activation='sigmoid')(decoded)

# Autoencoder Model
autoencoder = models.Model(input_img, output_img)

# Compile
autoencoder.compile(optimizer='adam', loss='binary_crossentropy')

# Summary
autoencoder.summary()

Train the Model
history = autoencoder.fit(
x_train, x_train,
epochs=10,
batch_size=256,
validation_data=(x_test, x_test)
)

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Evaluate & Test
decoded_imgs = autoencoder.predict(x_test)

Visualization
n = 5 # number of images to display
plt.figure(figsize=(10, 4))
for i in range(n):
# Original Image

```

```
ax = plt.subplot(2, n, i + 1)
plt.imshow(x_test[i].reshape(28, 28))
plt.title("Original")
plt.axis("off")

# Reconstructed Image
ax = plt.subplot(2, n, i + 1 + n)
plt.imshow(decoded_imgs[i].reshape(28, 28))
```

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```
plt.title("Reconstructed")
plt.axis("off")
plt.show()
```

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Conclusion:

Designed and trained a neural network to encode images into a lower-dimensional representation and then

decode them back to their original form. The visualization demonstrated the model's ability to reconstruct

images with reasonable fidelity, showcasing the effectiveness of autoencoders in learning efficient data

representations. This provides a solid foundation for further exploration into more complex image compression techniques.